Sensing at the edges: synthetic specimens for early contractile coordination

Ben Gaskin

University of Sydney, Australia
bgas0204@uni.sydney.edu.au

Abstract

How can a multicellular body coordinate adaptive behavior before any cell is specialized into sensor or effector roles? The skin brain thesis and its ASMO extension propose that such coordination arises from coupling generic contractile cells whose patterned activity is perturbed by what the body encounters. We construct synthetic specimens to investigate this regime: fixed 7×2 contractile bodies in which every cell runs the same continuous-time recurrent neural network with no imposed sensory or effector specialization. Evolved in a bilateral goal-cliff environment, these specimens explore, reverse at cliff edges, and reach goals, with mean native success of 89.6% (SD 12.4%) across eight selected specimens; while on longer, untrained distances performance approaches the 50% baseline of undirected commitment. Contrasting failure modes are consistent with deformation-mediated edge modulation and signaling-supported locomotor activity: removing deformation cuts native success to 47.9% and raises falls, while removing signaling impairs coordination with distinct stall and fall signatures. Morphological transfer preserves visible contractile locomotion but not the discrimination, consistent with calibration to morphology-dependent sensory and mechanical dynamics. One specimen, S4, shows the same selection regime admits another organization: under deformation ablation it reverses with no terrain-sensitive route into its controller, which runs open-loop with respect to the body and terrain while the driven body carries the reversal. Conditional coordination is thus realizable in a uniform contractile body before sensor–effector specialization—organized usually through deformation, but in S4 through driven body mechanics without a direction-setting controller signal.

Introduction

Single cells were already contractile, sensitive, and communicative long before multicellular bodies existed—competent organisms operating in, and responding to, their environments. When such cells came together, their environment changed: it was now largely composed of other cells with the same capacities, running the same programs. How these pre-existing competencies became the basis for coordinated multicellular behavior is not a question that accounts beginning from differentiated sensory and motor roles take up; such accounts presuppose those roles, starting with dedicated sensory cells that receive information from the world, intermediate networks that process it, and effector cells that act on the world in turn. The Braitenberg vehicle (Braitenberg, 1984) is the canonical minimal case, and Jékely's reconstruction of the earliest neural circuits as the efficient control of ciliary locomotion by a small number of specialized sensors (Jékely, 2011) is perhaps its sharpest recent articulation. Keijzer, van Duijn and Lyon (2013), however, argue that this is precisely the question that matters: how does a multicellular body come to behave as a unit at all?

Keijzer's skin brain thesis (SBT) and its development as the animal sensorimotor organization (ASMO) hypothesis (Keijzer, 2015) posit a pre-specialization regime in which coordination across a contractile body arises from the coupling of cells that are already, individually, contractile, sensitive, and communicative. The thesis's most provocative phrase—that the earliest such organisms operated "without input and output"—does not deny environmental sensitivity or action; it rejects treating early sensorimotor organization as a sequence of separable input and output stages. We operationalize one aspect of this proposal by imposing no specialized sensory or effector cell types. An extant analogue illustrating the relevant ingredients can be found in the choanoflagellates.

Choanoeca flexa is a choanoflagellate that forms multicellular sheets—clonally, aggregatively, or through mixed development depending on conditions (Brunet et al., 2019; Ros-Rocher et al., 2026). In response to a light-to-dark transition, these sheets undergo coordinated curvature inversion through the apical constriction of their constituent cells—a light-regulated collective behavior that dissociated cells do not exhibit (Brunet et al., 2019; Brunet, 2023). No cell is a dedicated sensor or effector: every cell senses light through the same rhodopsin–cGMP pathway and contracts through the same actomyosin machinery. Our starting point is whether such collective behavior can be shaped by selective pressures applied to the aggregate.

What becomes possible when generic cells—already capable of contractility, sensing, and local signaling—are cou-

© 2026 Ben Gaskin. Published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

pled within a multicellular body and placed under selection on the behavior of the whole? Using the methods of artificial life, we construct such synthetic specimens and ask what behavioral coordination they can be selected to produce, and what possibilities this establishes for coordination in clonal aggregates.

We evolve an artificial organism to study this regime under controlled conditions: a fixed contractile multicellular body—a voxel soft body based on those of EvoGym (Bhatia et al., 2021)—in which every cell runs the same continuous-time recurrent neural network (CTRNN; Beer, 1995, 2006). These cells have access only to local signals from bodily deformation and to the state of their immediate neighbors through a small number of signaling channels. Following Keijzer's ASMO, we treat deformation as a mechanotransductive form of sensing. Cells sense the mechanical effects of their own actions and their bodily interactions with neighbors and environment alike. As in *C. flexa*, no cell-type or parameter specialization is imposed: every cell has the same sensing and actuation channels and shares the same controller parameters, and any differentiation arises from position, body boundary, coupling, and signaling dynamics rather than from a labeled role.

To this end we draw upon the neural cellular automata introduced by Mordvintsev et al. (2020) as a model of morphogenesis, with complex multicellular organization emerging from local interactions among identical units. The hidden channels of these cells were framed there as being interpretable "as concentrations of some chemicals, electric potentials or some other signaling mechanism" (Mordvintsev et al., 2020). Where the original formulation used a stateless multi-layer perceptron, here we instead use a continuous-time recurrent neural network as our update rule. We do so not merely to provide a stateful controller, but further to emphasize this reading of NCAs in terms of an artificial chemistry: the recurrent dynamics embedded in our cells constitute the substrate of an evolvable chemistry, a coupled dynamical system whose influences are carved by selection rather than set down in advance, and which is embedded in a body such that the activities of these cells are simultaneously structured by and structure the body's interface with the environment (Dittrich et al., 2001).

We thus ask whether such an evolvable artificial chemistry—the shared logic each cell runs as part of a fixed contractile body—can, under selection on the behavior of the whole, produce ASMO-style coordination: behavior in which sensing and acting are one process and no cell is specialized. Each specimen, a set of cells in a fixed worm-like rectangle, is placed on an asymmetric plateau between a goal and a cliff. The terrain is mirrored and the distances varied, so that success requires the body to sense the ledge and on that basis reach the target. We use *discrimination* throughout in a behavioral sense—the body must produce different trajectories at the cliff and goal boundaries—with no internal classifier or representation assumed. The ecology is minimal by design: a fixed unilateral commitment does not suffice, and body-mediated edge sensing is favored. We then collect specimens from it, extend their tasks, and perturb their dynamics to map this space of possibilities for early coordination.

This paper offers three contributions. The first is substantive: a uniform, locally coupled contractile body, with no imposed sensory or effector specialization—every cell carries the same sensing, signaling, and actuation channels—can be selected for whole-body behavior *conditional* on what it senses, a minimal cliff-versus-goal discrimination. Across most selected specimens this behavior depends on deformation; S4 reveals a mechanically mediated alternative. This is the distinction from the neighboring morphological-computation line: Mertan and Cheney (2024) obtain cue-dependent behavior through reactive body materials and, separately, controller-free morphology-only classification (2025); here no subset of cells or materials is assigned a sensory or effector role—every cell shares the same sensing, signaling, actuation, and controller capacities, and the discrimination is selected into a uniform body. It carries de Wiljes et al. (2015)'s contractile-patterning lineage a step further, from the self-organization of patterning to its use in adaptive behavior. The second is analytic: through perturbation analyses we identify the discrimination's principal dependencies—the relations among locomotor drive, deformation, intercellular signaling, and body geometry—with contrasting failure modes consistent with signaling-supported locomotor activity and deformation-mediated edge modulation. The third is a limiting case: one specimen, S4, reverses at the edge with no terrain-sensitive route into its controller, the loss of support redirecting its driven body—an existence proof that the same regime admits more than one organization. Underpinning all three, we read the shared CTRNN-NCA as an evolvable, influence-level artificial chemistry: the means by which the biological question is made experimentally tractable, an alternative to the differentiable reaction networks of Mordvintsev et al. (2023) with tradeoffs we make explicit. Throughout, our object is the *space of possibilities* this regime admits rather than its typical behavior; the eight specimens map some of that space, and we defer how frequently each organization arises to future work.

Background

The skin brain thesis and animal sensorimotor coordination

Keijzer's ASMO (2015) develops the claim introduced above into a specific account of sensing. In an organization of generic contractile cells, sensing is not a separate capacity layered on top of coordination: it arises from the body's own patterned contraction being perturbed by what the body encounters. What the organism senses is constituted by what

its action produces in the cells that produce that action, so that acting and sensing are aspects of the same distributed process. Brembs (2021) develops a related active-dynamic view: behavior is not a response to stimulation but an intrinsically generated activity that the environment modulates. The specific claim our ecological task is designed to investigate is whether a contractile body, sensing through its own deformation, can on that basis produce adaptive directional behavior.

The cellular capacities this regime requires are not hypothetical. Actomyosin contractility is ancient and widespread across eukaryotes (Brunet, 2023; Guest et al., 2025), and mechanosensitive ion channels are conserved across eukaryotic lineages (Murthy et al., 2018), letting cells detect force and deformation through conformational changes in membrane proteins (Vogel and Sheetz, 2006). Multicellular coordination couples these pre-existing capacities rather than inventing new ones; our model's deformation channels abstract the same mechanosensitive capacity.

Following Keijzer (2015), a central computational question is whether such coordination is realizable at all—an old idea, traceable to Pantin's (1956) view of the animal as a contraction-grounded "behavior machine," now made precise. de Wiljes et al. (2015) address its *patterning* side, demonstrating that patterned contractile activity can self-organize across an excitable epithelium under plausible assumptions; they model a worm-like epithelium varied over numbers of cells and length–circumference proportions. The present paper takes the next step, reducing that geometry to a tractable two-dimensional contractile strip, embedding it in a body under a selection task, and selecting on behavior rather than on patterning alone.

Distributed controllers for 2D soft-body voxel robotics

The voxel soft-robot literature supplies the architectural precedents for our system. Medvet et al. (2020) introduced per-voxel neural controllers for voxel-based soft robots, contrasting centralized control—a single network reading sensors from across the body and writing actuation back to every cell—with distributed control, in which each cell carries its own small network and communicates only with its neighbors, and found the distributed case produces comparable locomotion. Nadizar et al. (2022) push this to a *uniform* neural cellular automaton in which every cell runs the same local rule with the same parameters, so that differentiation arises entirely from position and coupling—the technical form of what the preceding section describes in biological terms—and Nadizar et al. (2023) find stateful per-cell dynamics outperform stateless ones, allowing richer temporal coupling with the body. Related work has used a single NCA as a unified body–brain substrate for modular robots (Pontes-Filho et al., 2022) and evolved self-organizing embodied neural controllers for voxel soft robots (Pigozzi and Medvet, 2022). These uniform distributed-controller precedents principally study locomotion and its robustness; our contribution is to fix the morphology and select specifically for a cliff-versus-goal discrimination—behavior conditional on what the body senses—rather than locomotion alone.

We ask for something different in kind—situation-dependent rather than invariant behavior: behavior *conditional* on what the body senses, in which the action taken depends on a discrimination the body makes about its situation. Flat-ground locomotion does not force sensing to matter, because open-loop pattern generation suffices; our task is built so that it does, requiring the body to tell a cliff edge from a goal edge and act differently at each. A complementary line already treats the body as the locus of such function rather than as an aid to control: from the body-as-reservoir tradition, in which passive morphological dynamics serve an abstract control layer (Hauser et al., 2011), to the controller-free morphological computation of Mertan and Cheney (2024, 2025), in which voxel bodies with fixed, simple cell behaviors produce cue-dependent behavior—and classify their environment—with no controller at all. Our cliff-versus-goal task asks for something closer to what they call *morphological cognition*: different behavior in different situations, here in a uniform body with no cell assigned a sensory or effector role.

NCA controllers as an evolvable artificial chemistry

The introduction framed the CTRNN-NCA as an evolvable artificial chemistry; here we make the reading precise and bound it. NCA cell states have been read as the physiological status of biological cells, with the shared update rule as a model of intracellular regulation (Hartl et al., 2026). We take this one step further, from analogy toward an influence-level artificial chemistry, while being clear about what kind. The reading is *influence-level*, not reaction-level: the recurrent weight matrix says which intracellular variables influence which others, and with what sign and strength, but it is not a stoichiometric reaction graph (Dittrich et al., 2001). Each cell is a local reactor running the same dynamics under different local conditions, and evolution carves the influence graph the ecology demands. We interpret the learned bias as constitutive basal activity, providing tonic endogenous drive under deterministic zero initialization.

The most direct comparison is to Mordvintsev et al. (2023), who program differentiable reaction networks—mass-action chemistries tuned by gradient descent—and show that coupled chemical cells can compute collective functions. Our model makes a different choice. Reaction-level models preserve stoichiometric structure, whereas the CTRNN provides a compact continuous-time state space whose signed influences, tonic activity, and relaxation times can be shaped by evolutionary search. We use it for exactly these properties: its continuous-time dynamics, its small

evolvable parameterization, and its natural reading as an influence-level chemistry. We make no claim that this fixed twelve-variable controller realizes an arbitrary reaction-level chemistry.

Methods

Our experimental system has four components: a simulated physical environment with a fixed morphology, a distributed controller, an ecological task, and an evolutionary search over controller parameters. The simulator, JAX-EvoGym, is a pure-JAX reimplementation of Evolution Gym (Bhatia et al., 2021),[1] extended with physically motivated ground friction, dual-mode deformation sensing (absolute and compliance, defined below), and a new contractile cell type—comparable to Evolution Gym's soft voxel, but combining it with independent horizontal and vertical actuation so that one cell can contract along either axis. One setting departs from Evolution Gym's defaults and matters for what follows: actuator convergence is slowed relative to the default, with the value and calibration given in the supplement.

Body and actuation

The body is a fixed 7×2 rectangle of 14 identical contractile cells—an undifferentiated contractile sheet in which every cell runs the same program. We take this fixed two-dimensional strip as a tractable reduction of the worm-like epithelial geometry de Wiljes et al. (2015) study, holding the body constant during evolution so as to isolate the question of coordination from that of morphology search. As we report below, the morphological-transfer assay later probes the consequences of holding the body fixed, and relaxing it is a natural direction for further work. Each cell actuates by setting a target rest-length multiplier within a fixed band, contracting or extending its spring symmetrically about its reference length. Episodes run for 3000 environment steps.

CTRNN-NCA controller

Each cell maintains 12 private intracellular state variables, updated by its CTRNN, and writes to 6 signaling channels visible to its 3×3 neighborhood, in the manner of a neural cellular automaton (Mordvintsev et al., 2020)—though with a richer perception than the original NCA, detailed below (Figure 1). All cells share the same learnable parameters. Each state variable relaxes toward a weighted combination of the cell's intracellular states, its perception inputs, and a learned bias, on a learned per-variable time constant (full dynamics in the supplement); the action and signaling outputs themselves carry no bias term. Under the deterministic zero reset used here, a controller with no input would otherwise rest at its quiescent fixed point; the learned state biases provide the tonic endogenous activity that initiates non-zero dynamics and seeds the oscillation underlying locomotion, while the learned time constants give the intracellular variables distinct relaxation scales.

Each cell perceives, by identity, its own deformation and two command-feedback channels carrying its previous action outputs, and perceives its neighbors' signaling through a bank of spatial filters (Sobel-x, Sobel-y, Laplacian, identity) over the 3×3 neighborhood (full perception detail in the supplement). The deformation channels encode two modes along each axis, absolute and compliance. Absolute deformation—a cell's spring state relative to its reference configuration—lets the cell sense the mechanical consequences of its own and its neighbors' action (cf. Vogel and Sheetz, 2006). Compliance deformation instead measures the spring state relative to the cell's own current actuation goal; we include it because, at the soft-cell stiffnesses used here, absolute excursions are small relative to the scale at which the controller reads them, and compliance supplies a better-conditioned signal. We adopt it for tractability and flag the conceptual cost: compliance is a command-relative proprioceptive signal that exposes the discrepancy between commanded and realized deformation, a stronger assumption than the bare deformation sensing the framing otherwise rests on. We accept that it supplies the controller with something a body could in principle have to evolve for itself, our interest being the dynamics of whole-body coordination rather than the evolution of that command-referenced signal. Each cell outputs two actuation commands and six signaling updates, and all cells update synchronously from a deterministic zero reset, with no noise or random initialization. The bias-seeded oscillation is uniform across cells; boundary effects in the spatial filters spatialize this initially uniform activity, providing a body-fixed locomotor frame with no absolute positional coordinate. The cliff is never a labeled sensory channel. In intact runs, terrain affects controller state only through contact-driven deformation; when deformation is removed, support loss still perturbs the body but has no route back into the controller.

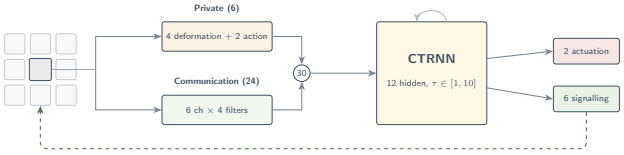


Figure 1: Controller architecture for a single cell. Perception splits into 6 private channels and 6 communication channels, yielding a 30-dimensional input. The CTRNN maintains 12 intracellular state variables with learned time constants and produces 8 outputs: 2 actuation channels mapped to rest-length multipliers and 6 signaling-channel updates written to the shared grid (dashed arrow).

[1] Animations, the full model specification, and supplementary analyses are available in the web supplement, together with links to the JAX-EvoGym codebase: [bengaskin.com/sensing-edges].

Ecological task



Figure 2: The bilateral goal-cliff environment. **(A)** A 7×2 body on the native core terrain: a 12-cell central plateau flanked by a goal block (left) and a cliff (right), shown for distance pair (2, 1). **(B)** An extended terrain with greater goal and cliff distances, used for generalization testing. **(C)** The symmetric double-cliff terrain (plateau width 9), with cliff drop-offs on both sides and no goal.

The body is placed on a 12-cell-wide central plateau flanked by a raised goal block on one side and a cliff drop-off on the other (Figure 2). Three asymmetric distance pairs $(d_{\text{goal}}, d_{\text{cliff}}) \in \{(2,1),(4,2),(6,3)\}$ produce 6 terrain configurations when goal and cliff sides are mirrored across episodes, yielding 6 evaluation episodes per candidate. The mirroring defeats fixed unilateral commitment and rewards edge-dependent body–environment dynamics: the body must use its own state to distinguish the two edges, commit to a direction, and reverse upon encountering the cliff. The cliff is also placed closer than the goal, an asymmetry that discourages a hedged or weakly committed gait. On a mirrored bilateral terrain, committing to a single direction without sensing succeeds exactly when the chosen direction happens to be correct, giving the 50% mirror-pair baseline of a fixed undirected commitment policy against which sensing-dependent success must be read. Our interest is in the *capacities* this scenario can produce, not in the evolutionary route to them; the fitness function is accordingly a shaped, constructive search instrument rather than a model of sparse natural selection. Per-episode fitness rewards, in sequence, exploration toward the cliff side, reversal once wrong-side progress is meaningful (gated so the reversal term counts only then), and approach to and contact with the goal, while penalizing falls. Candidate fitness is the mean episode score plus a bonus per mirrored pair solved (both the left and right variants of a distance pair). Distances are in terrain grid cells, not measured from the center of mass. We score a goal hit when any body point touches the goal block, success as a goal hit without a fall, a fall as death-plane contact, and a stall as neither by the step limit; on double-cliff terrains survival means not falling and is not goal success.

Evolution and selection of specimens

Specimens are evolved with sep-CMA-ES (Ros and Hansen, 2008) emitters, implemented via evosax (Lange, 2023), within the CMA-MAE framework (Fontaine and Nikolaidis, 2023), over a MAP-Elites archive (Mouret and Clune, 2015) indexed by three behavioral descriptors—goal-hit count, fall count, and completion time. Across five evolutionary runs we retained the top eight archive specimens with a cut-off of 4 successful runs out of 6 possible trials; selection preceded all ablation and mechanism analyses, and all results below come from a separate deterministic post-run assay. The eight are not independent evolutionary replicates, and the deterministic terrain episodes are not stochastic trials; we treat the specimen as the unit of analysis, reporting descriptive statistics across the eight. S4's distinctive status was identified only afterwards, so its mechanistic analysis is post hoc.

Post-run assays

Selected specimens are subjected to several families of post-run assay. All evaluations are deterministic: for a given terrain configuration the controller produces the same behavior, with hidden states initialized to zero; each episode corresponds to a single terrain configuration, not a repeated trial. Native core evaluation comprises 6 episodes per specimen (3 distance pairs $\times$ 2 sides). Generalization is tested on extended distance pairs $(8,4)$ and $(10,5)$, and on symmetric double-cliff terrains (cliff widths 2, 4, and 6), the latter as a survival stress assay in which both sides present a cliff and no goal exists. Deformation ablations compare four conditions—intact, absolute-only, compliance-only, and deformation-off—isolating the contribution of each deformation mode. Signaling ablations selectively disable spatial filters: Sobel-x, Sobel-y, Laplacian, identity, and all communication. Zero-shot body transfer tests each controller on 6×2, 8×2, and 7×3 morphologies.

Results

We report descriptive statistics across the eight selected specimens; each specimen contributes a single rate per condition, and dispersion is the standard deviation across specimen-level rates. Where one specimen (S4) drives between-specimen spread, we report results with and without it.

Native performance

On the native core evaluation, specimens achieved a mean success rate of 89.6% $\pm$ 12.4% (43 of 48 episodes) with a fall rate of 2.1% $\pm$ 5.9%. Performance was perfect on the two shorter pairs (2,1) and (4,2); all failures occurred on the hardest pair (6,3), where success fell to 68.8% $\pm$ 37.2%. Qualitatively, specimens exhibit rhythmic contractile waves that propagate along the body, producing directed locomotion; upon encountering a cliff edge, the body's deformation pattern changes and the specimen reverses direction, committing to the goal side. Specimens varied in the strength of their directional commitment—the strongest committed cleanly and reversed sharply, while others vacillated, exploring tentatively before settling. The native rate is a post-selection capacity measure; the quantities we rely on are the

within-specimen paired changes under perturbation reported below.

Generalization and robustness

Extended terrains test whether the *discrimination*, not merely the locomotion, transfers to distances not seen during training. Success was lower and heterogeneous: 56.2% ± 32.0% on (8,4) (9 of 16 episodes) and 68.8% ± 25.9% on (10,5) (11 of 16), with a combined extended success of 62.5% (20 of 32). On these mirrored terrains, 50% is the mirror-pair baseline of committing to a single direction without sensing. The combined extended performance does not clearly exceed this baseline, and it is concentrated in a minority of specimens: excluding S4, combined extended success falls to 57.1%. Visible contractile locomotion persists—specimens still produce contractile waves and move on the longer terrains—but the cliff-versus-goal discrimination transfers only partially: beyond the trained distance range it remained above the unilateral-policy baseline in some specimens, but heterogeneously and not reliably across the cohort.

The double-cliff assay, in which both edges drop away and no goal exists, tests whether edge avoidance generalizes beyond the trained cliff-then-goal sequence to a situation in which either direction can lead to support loss; it separates the specimens as a survival stress test rather than a goal task. Reported as a survival matrix, only one specimen survived at width 2 (1/8), one at width 4 (1/8), and two at width 6 (2/8); two specimens survived at least one width, and one of these—S4—survived all three (Figure 3). We report these as counts rather than as a mean, since the distribution is degenerate.

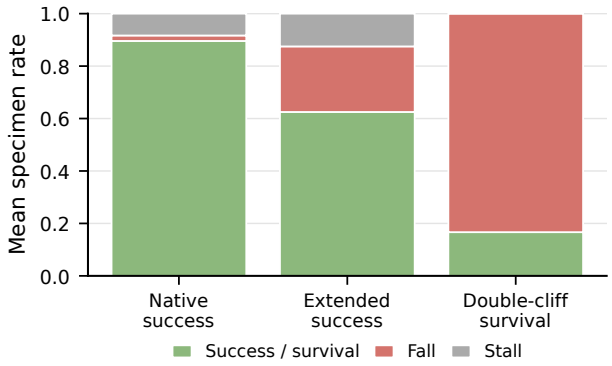


Figure 3: Native performance, extended generalization, and double-cliff survival across the eight selected specimens. Bars show means of specimen-level outcome rates.

Deformation ablations

Removing all deformation feedback dropped native success to 47.9% ± 27.4% and raised falls to 35.4%—a paired within-specimen change of −41.7 percentage points, and a rise in falls of more than thirty points over intact controls (Figure 4). Retaining either channel preserved some performance, with somewhat greater retention under compliance-only sensing (60.4% success, −29.2 pp) than absolute-only (54.2%, −35.4 pp). These paired deltas, computed within each specimen against its own intact baseline, are the summary we rely on; excluding S4, the deformation-off cohort falls further, to 40.5% success. For most of the eight selected specimens, successful edge-contingent performance depends strongly on the deformation input: removing it largely abolishes edge-contingent performance and cliff safety.

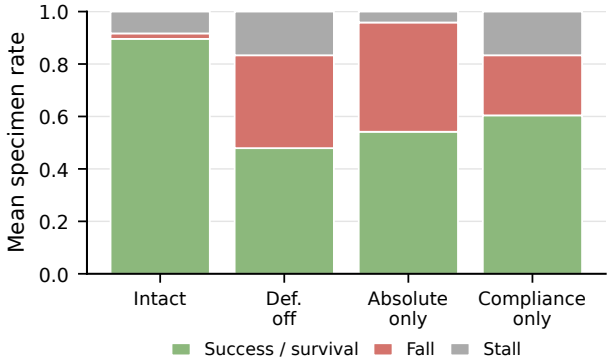


Figure 4: Native deformation-ablation outcomes across the eight selected specimens. Bars show specimen-level mean outcome rates; each condition comprises six deterministic native terrain episodes per specimen.

Signaling ablations

Each spatial-filter ablation produced a distinct behavioral impairment, with both the magnitude and the failure mode varying across filters (Figure 5). Disabling Sobel-x produced the sharpest single-channel impairment—success 18.8%, a paired change of −70.8 pp—with failures split between falling (31.2%) and stalling (50.0%). Disabling all communication dropped success to 27.1% (−62.5 pp); here the dominant failure mode was stalling (47.9%) rather than falling, so that without intercellular signaling specimens tended to freeze rather than run off cliffs. The remaining single-filter ablations were intermediate and comparable in magnitude—Laplacian −54.2 pp, Sobel-y and identity each −50.0 pp—with distinct mixtures of falls and stalls. Although the failure patterns are suggestive, the ablations do not isolate filter-specific functions; we therefore interpret the filters collectively as a signaling interface.

Morphological variation

Zero-shot transfer to non-native bodies was tested on native core terrains for all eight specimens. On the native 7 × 2

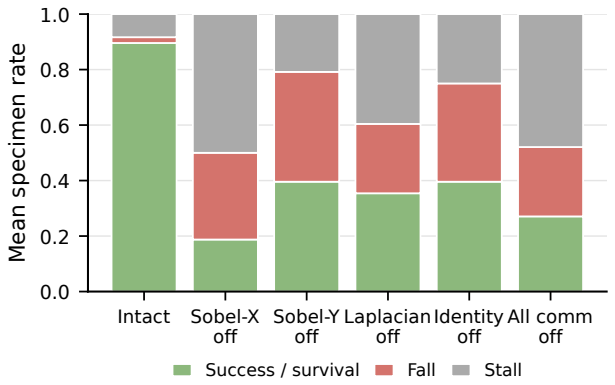


Figure 5: Native signaling-ablation outcomes across the eight selected specimens. Bars show specimen-level mean outcome rates; each condition comprises six deterministic native terrain episodes per specimen.

body these specimens succeed at 89.6%; on a 6×2 body success collapses to exactly $50.0\% \pm 0.0\%$ for every specimen, and to 39.6% (8×2) and 41.7% (7×3), with falls rising. That every specimen lands at precisely the undirected baseline on the 6×2 body is striking: visible contractile locomotion persists—specimens still produce contractile waves and move—but the cliff-versus-goal discrimination does not. Edge-contingent task performance is thus morphology-specific even where locomotion persists; we return in the discussion to why that might be.

Variation across specimens, and a limiting case

The ablation rates reported above are cohort means, and the dispersion around them is large and informative. A closer look at the per-specimen degradation shows that the effect of a given ablation is far from uniform: under deformation removal some specimens lose cliff safety almost entirely while others are barely affected, and the deformation effect in particular ranges across the cohort from near-total dependence to none. This spread is itself a finding. The same task, body, and controller substrate do not converge on a single coordination strategy; they produce a range of them, differing in how drive, deformation, and intercellular signaling are combined.

One specimen, S4, lies at the far edge: it is perfect on native and extended terrains, survives all three double-cliff widths, and—alone among the eight—keeps full success when deformation is removed, while collapsing to a complete stall when Sobel-x or all communication is removed. Where most of the other specimens sense the cliff through deformation, S4 reverses at the edge with no terrain-sensitive route into its controller. This is surprising, and we treat it as a single well-characterized case rather than as typical of the regime.

Three facts make the case. First, the reversal tracks the physical edge, and an exact control isolates the moment of support loss. Across an extended deformation-off sweep with terrain width held constant and cliff distance varied from 1 to 30 cells, S4 releases and returns from the edge on all thirty configurations. Here 28 of 30 reached the goal and none fell, with the release time increasing monotonically with cliff distance (Spearman $\rho = 1.0$): the reversal thus tracks the edge across distances rather than any fixed internal clock. Because only the cliff distance varies, the body streams of any two trials are bitwise identical up to the earlier of their two release frames, and the signaling streams remain so throughout.

Second, the systematic return depends on the loss of support, acting through the driven body, not on a timed command. With the deformation input removed, S4's controller is open-loop with respect to the body and terrain, so a single command stream underlies every cliff distance and only the timing of support loss differs; because the traces are bitwise identical until release, the small directional excursions in that shared gait are features of the rhythm rather than distance-specific events. On the real cliff, support loss is followed by a sustained away-from-cliff regime that fills most of the post-release run; replaying the same commands on a long supported terrain reproduces only brief directional excursions, not the systematic sustained return. The return is delayed, the body holding its cliffward phase for some hundreds of steps before it takes hold.

Third, the analyzed controller streams carry no detectable information about the body's subsequent direction, while the body's mechanical streams do. In held-out analyses, the mechanical streams (i.e., kinematics and spring state) distinguished subsequent sustained away- from toward-moving phases, whereas the analyzed controller-state and command streams remained at chance. The loss of support redirects the driven body while the controller runs on its rhythm. That the drive itself is signaling-borne is a separate matter: remove the deformation input and the Sobel-x signaling filter together and S4 stalls (0 of 6), abolishing the locomotor drive. Deformation, when present, steadies the approach and the initial commitment rather than producing the reversal.

Discussion

Uniform distributed-controller work establishes locomotion and its robustness, while morphological-computation work establishes cue-dependent behavior through reactive materials; we show that edge-contingent behavior can also evolve in a uniformly composed, locally coupled contractile body with no cell assigned a sensory or effector role, suggesting that the regime SBT and ASMO describe is computationally plausible. The selected specimens support claims about capability not optimality or frequency; our object is the *space of possibilities* the regime admits rather than its tendency.

Sensing, signaling, and their limits

We have argued that the discrimination rides on an intrinsic locomotor drive that sensing modulates. The drive appears to be sustained by intercellular signaling: when removed, coordination breaks down and stalling becoming the dominant failure. For most selected specimens, deformation input supplies the edge-sensitive modulation: remove it and specimens keep moving but lose the discrimination, their approach growing hesitant and their success falling toward the undirected baseline, with the within-specimen paired deltas anchoring this for most of the eight. The two are complementary: the task requires both sustained locomotor activity and edge-sensitive modulation. This resembles the active-dynamic relation Brembs (2021) describes: an intrinsic ongoing activity modulated rather than instructed by what the body encounters, with the controller's bias-seeded drive as the intrinsic activity and sensing as its modulation.

The per-specimen variation, however, showed the deformation effect ranging from total dependence to none, and its limiting case sharpens the claim S4 reverses at the cliff with no terrain-sensitive route into its controller: release produces no direction-setting change in its controller state or commands, while the loss of support redirects the driven body. The deformation input to the controller is thus not strictly *necessary* to the discrimination: the regime admits a route that does without it, even as physical deformation and the support loss remain integral to the mechanics. S4 occupies a middle position between controller-mediated sensing and the controller-free morphological computation of Mertan and Cheney (2024, 2025). The environmental event does not act through a designated reactive material; it perturbs the dynamics of a uniformly composed, actively driven body. The edge remains a mechanical cause without becoming direction-setting information for the controller—a distinct organization of environment-contingent behavior whose generality across the regime remains open.

Why the discrimination is body-specific

The transfer assay observes that the discrimination is body-specific where the locomotion is not: on a different body the cliff-versus-goal behavior collapses to baseline while the contractile waves persist. S4 suggests why. There the reversal is carried by the body's own mechanics—the forces between voxels, the loss of support at the edge—acting as a coordinative medium in their own right; if a milder version holds across the cohort, with deformation-reading leaning partly on the same geometry-dependent mechanics, the discrimination should be body-specific in just the way we see, since a body of different proportions produces different mechanical signatures for the same terrain feature, and a controller-and-signaling system calibrated to one geometry reads them poorly. This is interpretation, not demonstration—we have direct evidence only that S4 reverses through its mechanics—but it makes the body-specificity intelligible, and casts the body as part of the sensory interface rather than a passive output device. A route forward follows from it: plastic or totipotent controllers (Ferigo et al., 2025) offer a direct test of whether online adaptation can restore edge-contingent performance across altered bodies, where morphological change often degrades an evolved controller (Medvet and Rusin, 2023)—most valuable once bodies are themselves grown through morphogenetic stages (Pontes-Filho et al., 2022), eventually with differentiated sensory cells (Ferigo et al., 2021).

Conclusion

We asked whether a uniform contractile body could evolve situation-dependent behavior without imposed sensory or effector roles. Under selection in a bilateral goal-cliff environment, our specimens—fixed contractile bodies in which every cell runs the same evolvable artificial chemistry, sharing only local signaling—produced support-contingent reversal and goal-reaching behavior. In most cases, however, this edge-contingent performance transferred to new distances and symmetric-risk scenarios only partially. While ablations indicate that intercellular signaling sustains locomotor drive while deformation supplies the edge-sensitive modulation, the deformation effect varied sharply across specimens. The limiting case, S4, marks an alternative: under ablation its controller supplies an ongoing drive while the loss of support redirects the body, without the edge appearing as input. Together these results carry de Wiljes et al. (2015)'s account of contractile coordination forward and establish this synthetic framework as a productive substrate for studying the pre-specialization regime Keijzer (2015) describes.

Acknowledgements

This research was supported by an Australian Government Research Training Program (RTP) Scholarship. This project was supported in its programming and editing by Claude Code and Codex. All mistakes, however, are my own.

References

Beer, R. D. (1995). On the dynamics of small continuous-time recurrent neural networks. *Adaptive Behavior*, 3(4):469–509.

Beer, R. D. (2006). Parameter space structure of continuous-time recurrent neural networks. *Neural Computation*, 18(12):3009–3051.

Bhatia, J. S., Jackson, H., Tian, Y., Xu, J., and Matusik, W. (2021). Evolution Gym: A large-scale benchmark for evolving soft robots. In *Advances in Neural Information Processing Systems*, volume 34, pages 2201–2214.

Braitenberg, V. (1984). *Vehicles: Experiments in Synthetic Psychology*. MIT Press, Cambridge, MA.

Brembs, B. (2021). The brain as a dynamically active organ. *Biochemical and Biophysical Research Communications*, 564:55–69.

Brunet, T. (2023). Cell contractility in early animal evolution. *Current Biology*, 33(18):R966–R985.

Brunet, T., Larson, B. T., Linden, T. A., Vermeij, M. J. A., McDonald, K., and King, N. (2019). Light-regulated collective contractility in a multicellular choanoflagellate. *Science*, 366(6463):326–334.

de Wiljes, O. O., van Elburg, R. A. J., Biehl, M., and Keijzer, F. A. (2015). Modeling spontaneous activity across an excitable epithelium: Support for a coordination scenario of early neural evolution. *Frontiers in Computational Neuroscience*, 9:110.

Dittrich, P., Ziegler, J., and Banzhaf, W. (2001). Artificial chemistries—a review. *Artificial Life*, 7(3):225–275.

Ferigo, A., Iacca, G., and Medvet, E. (2021). Beyond body shape and brain: Evolving the sensory apparatus of voxel-based soft robots. In *Applications of Evolutionary Computation (EvoApplications 2021)*, volume 12694 of *LNCS*, pages 210–226. Springer.

Ferigo, A., Iacca, G., Medvet, E., and Nadizar, G. (2025). Totipotent neural controllers for modular soft robots: Achieving specialization in body–brain co-evolution through Hebbian learning. *Neurocomputing*, 614:128811.

Fontaine, M. C. and Nikolaidis, S. (2023). Covariance matrix adaptation MAP-annealing. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '23)*, pages 456–465. ACM.

Guest, S. L., Velle, K. B., Jacques, S. M., Park, Y., Man, J., Titus, M. A., and Fritz-Laylin, L. K. (2025). Myosin 2 drives actin contractility in fast-crawling species outside of the amorphean lineage. *Current Biology*, 35(23):5696–5708.e6.

Hartl, B., Levin, M., and Pio-Lopez, L. (2026). Neural cellular automata: Applications to biology and beyond classical AI. *Physics of Life Reviews*, 56:94–108.

Hauser, H., Ijspeert, A. J., Füchslin, R. M., Pfeifer, R., and Maass, W. (2011). Towards a theoretical foundation for morphological computation with compliant bodies. *Biological Cybernetics*, 105:355–370.

Jékely, G. (2011). Origin and early evolution of neural circuits for the control of ciliary locomotion. *Proceedings of the Royal Society B*, 278(1707):914–922.

Keijzer, F. (2015). Moving and sensing without input and output: Early nervous systems and the origins of the animal sensorimotor organization. *Biology & Philosophy*, 30(3):311–331.

Keijzer, F., van Duijn, M., and Lyon, P. (2013). What nervous systems do: Early evolution, input–output, and the skin brain thesis. *Adaptive Behavior*, 21(2):67–85.

Lange, R. T. (2023). evosax: JAX-based evolution strategies. In *Companion Conference on Genetic and Evolutionary Computation (GECCO '23 Companion)*, pages 659–662. ACM.

Medvet, E., Bartoli, A., De Lorenzo, A., and Fidel, G. (2020). Evolution of distributed neural controllers for voxel-based soft robots. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference (GECCO '20)*, pages 112–120. ACM.

Medvet, E. and Rusin, F. (2023). Impact of morphology variations on evolved neural controllers for modular robots. In *Artificial Life and Evolutionary Computation (WIVACE 2022)*, volume 1780 of *CCIS*, pages 266–277. Springer.

Mertan, A. and Cheney, N. (2024). No-brainer: Morphological computation driven adaptive behavior in soft robots. In *From Animals to Animats 17 (SAB 2024)*, pages 81–92.

Mertan, A. and Cheney, N. (2025). Morphological cognition: Classifying MNIST digits through morphological computation alone. In *Proceedings of the 2025 Artificial Life Conference (ALIFE 2025)*. Paper 64.

Mordvintsev, A., Randazzo, E., and Niklasson, E. (2023). Differentiable programming of chemical reaction networks. arXiv:2302.02714.

Mordvintsev, A., Randazzo, E., Niklasson, E., and Levin, M. (2020). Growing neural cellular automata. *Distill*, 5(2):e23.

Mouret, J.-B. and Clune, J. (2015). Illuminating search spaces by mapping elites. arXiv:1504.04909.

Murthy, S. E., Dubin, A. E., Whitwam, T., Jojoa-Cruz, S., Cahalan, S. M., Mousavi, S. A. R., Ward, A. B., and Patapoutian, A. (2018). OSCA/TMEM63 are an evolutionarily conserved family of mechanically activated ion channels. *eLife*, 7:e41844.

Nadizar, G., Medvet, E., Nichele, S., and Pontes-Filho, S. (2022). Collective control of modular soft robots via embodied spiking neural cellular automata. arXiv:2204.02099.

Nadizar, G., Medvet, E., Nichele, S., and Pontes-Filho, S. (2023). An experimental comparison of evolved neural network models for controlling simulated modular soft robots. *Applied Soft Computing*, 145:110610.

Pantin, C. F. A. (1956). The origin of the nervous system. *Pubblicazioni della Stazione Zoologica di Napoli*, 28:171–181.

Pigozzi, F. and Medvet, E. (2022). Evolving modularity in soft robots through an embodied and self-organizing neural controller. *Artificial Life*, 28(3):322–347.

Pontes-Filho, S., Walker, K., Najarro, E., Nichele, S., and Risi, S. (2022). A single neural cellular automaton for body-brain co-evolution. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22)*. ACM.

Ros, R. and Hansen, N. (2008). A simple modification in CMA-ES achieving linear time and space complexity. In *Parallel Problem Solving from Nature (PPSN X)*, pages 296–305. Springer.

Ros-Rocher, N., Reyes-Rivera, J., Horo, U., Combredet, C., Foroughijabbari, Y., Larson, B. T., Coyle, M. C., Houtepen, E. A. T., Vermeij, M. J. A., Steenwyk, J. L., and Brunet, T. (2026). Clonal-aggregative multicellularity tuned by salinity in a choanoflagellate. *Nature*, 651:974–985.

Vogel, V. and Sheetz, M. (2006). Local force and geometry sensing regulate cell functions. *Nature Reviews Molecular Cell Biology*, 7(4):265–275.